Auditing Distributional Supervision for Learned Judges with Statistic-Preserving Label Randomization

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Abstract

Distributional supervision for learned judges is often interpreted as teaching how humans disagree. We audit the predictive consequences of replacing training histograms with donor-derived targets that retain either their mode or their mode and mean, and we report the intervention's support and total-variation dose. Across ten ordinal panels with frozen-feature heads, exact mode-and-mean-preserving replacement changed average RPS by -0.0024 to $+0.0006$, although several panels had low changed fractions or small perturbations. In two end-to-end Qwen2.5-7B LoRA blocks with Chen-style distributional training, mode-and-mean-preserving replacement of 69–83 % of training targets gave raw contrasts of -0.0001 [-0.0011, $+0.0011$] on SummEval coherence and $+0.0000$ [-0.0009, $+0.0009$] on POPQUORN offensiveness, whereas mode-only replacement increased RPS by $+0.0120$ [$+0.0086$, $+0.0155$] and $+0.0029$ [$+0.0009$, $+0.0050$], with positive estimates in all nine observed seed–fold cells on each dataset; the POPQUORN intervals condition on its finite rater panel, and under a rater-aware bootstrap its mode-only interval includes zero. In a complementary nominal construction, scrambling residual allocation while retaining modal class and mass increased KL on the three-class ChaosNLI MNLI and SNLI panels by 0.016–0.029 and 0.032–0.052; because the retained statistics, scoring rule, datasets and label structures differ, this comparison is descriptive. These learner-, dose- and panel-specific results do not establish that full disagreement distributions are generally unnecessary, nor that learned judges generally recover the full distribution of human disagreement.

1 Introduction

Learned judges — language models fine-tuned to score text — are increasingly trained on the full distribution of human ratings rather than on a single aggregated label. A rating histogram records information discarded by a single aggregate label, including the observed concentration of ratings and which neighbouring levels received mass; whether these empirical differences generalise to new raters is an empirical question. Prior work has reported lower proper-scoring loss from distributional objectives on several benchmarks (Chen et al., 2025; Singh et al., 2025; Kohli, 2026), but those comparisons do not establish which properties of the training histograms drive the improvement. Mode and mean summarise location; after fixing them, the remaining degrees of freedom include dispersion, skew, and allocation across the other levels. The broad claim that a model learns disagreement attributes predictive value to some part of that remainder.

We propose an intervention audit that separates these components without proposing a new training objective. For each training item, we assign a *donor* histogram from another training item within either a mode cell or a mode-and-mean-bin cell. For the exact variant, we then KL-project the donor to the recipient's exact mode and mean. LoRA arms share the data partition and configured optimizer seed; heads arms share the item/annotation split and master RNG-stream index but use arm-specific initialisation and minibatch seeds, so their contrasts are split-matched, not optimizer-seed matched. A contrast estimates the average predictive consequence of the realised target replacement, conditional on the learner, split, optimizer realisation, assignment rule and intervention dose; it does not identify the intrinsic value of the removed information. Support is the fraction of items in cells containing at least two histogram classes and is an upper bound on the changed fraction. We report support, realised changed fraction, TV among changed items and TV over all items, because these quantities delimit the interpretation of small observed contrasts, especially on panels with three to five ratings per item.

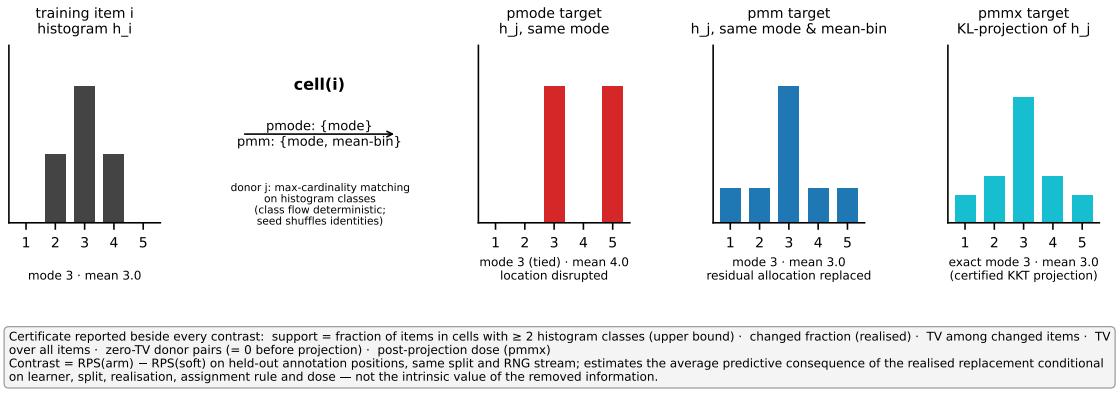


Figure 1: Statistic-preserving donor replacement (schematic; illustrative histograms). For a training item with histogram h_i, a donor histogram from another training item is assigned within a mode cell (pmode) or a mode-and-mean-bin cell (pmm); the exact variant (pmmx) KL-projects the donor to the recipient's mode and mean. The certificate reported beside every contrast lists support, realised changed fraction, TV among changed items, TV over all items and the zero-TV donor-pair count (0 before projection). Constrained donor construction does not imply statistical independence between original and replacement targets; the class-level assignment flow is deterministic given class sizes and the seed shuffles item identities within it.

We apply the audit in two settings. Frozen-feature heads trained on features from a 7B instruction-tuned model give breadth and repeated optimisation across ten ordinal panels (code readability, summarisation, helpfulness, conversational safety, offensiveness, politeness, sarcasm, emotional valence) and three nominal panels (ChaosNLI MNLI and SNLI as substantive three-class tests, plus binary αNLI as a zero-dose implementation check), and allow positive controls with injected shape. LoRA-fine-tuned judges provide a separate end-to-end study with Chen-style distributional training on two ordinal datasets — SummEval coherence (persistent experts held out) and POPQUORN offensiveness (annotation positions held out) — each with three joint optimizer–assignment realisations on one split; the two settings are reported side by side and are not assumed interchangeable.

Our findings are deliberately scoped. (i) Exact mode-and-mean-preserving replacement produced small average RPS changes on every ordinal head panel and in both LoRA blocks: -0.0001 $[-0.0011, +0.0011]$ on SummEval and $+0.0000$ $[-0.0009, +0.0009]$ on POPQUORN. (ii) Mode-only replacement, which additionally disrupts item-specific location, produced positive raw-RPS contrasts in all nine observed seed-by-fold cells on each LoRA dataset and positive observed contrasts on several head panels, but not uniformly across panels or pipelines; its smaller POPQUORN estimate ($+0.0029$ versus $+0.0120$) is a descriptive cross-dataset comparison, not a tested explanation. (iii) On the nominal ChaosNLI MNLI and SNLI panels, scrambling residual allocation while retaining the modal class and mass increased observed mean test KL at every tested annotation count. (iv) No constructed target had the lowest observed RPS on every ordinal panel. These results do not support a general claim that distributional judge training recovers the full disagreement distribution; the effects varied across tasks and were consistent with a role for label geometry, but the ordinal–nominal comparison is descriptive, and the dose certificates delimit what any single panel can say.

2 Related work

Distributional supervision and learned judges. Label distribution learning treats the per-item distribution over labels as the prediction target (Geng, 2016). For LLM judges, Chen et al. (2025) align the judge's output distribution with the empirical human distribution through a hybrid KL and cross-entropy objective with target-distribution adversarial perturbation; our SummEval LoRA block audits this objective, and the frozen-feature heads implement the same target interventions under a separate KL-trained learner. Post-hoc alternatives align a frozen judge's distribution with human distributions after the fact (NAPHA; Steindl et al., 2026). Elangovan

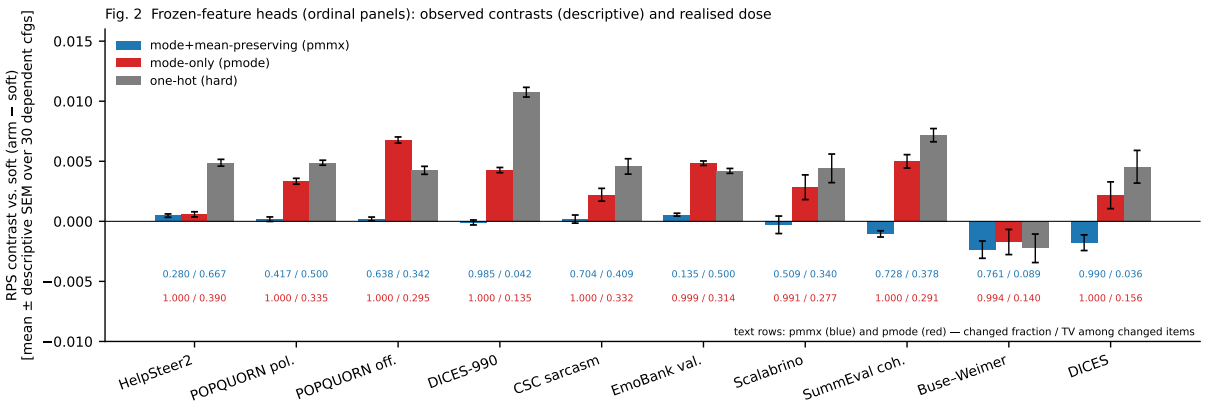


Figure 2: Frozen-feature heads: observed RPS contrasts (arm − soft) for exact mode-and-mean-preserving replacement (pmmx), mode-only replacement (pmode) and one-hot targets (hard) on ten ordinal panels. Bars are means and error bars descriptive SEMs over 30 dependent configurations (5 RNG streams × 3 item splits × 2 annotation-position splits); they are not inferential intervals, and the figure supports only descriptive average contrasts across the tested panels, splits and training realisations. Text under each panel gives the realised post-projection changed fraction and TV among changed items for pmmx (blue) and pmode (red).

et al. (2025) and Guerdan et al. (2025) evaluate judges against uncertain or indeterminate human ratings rather than training on them. To our knowledge, these works do not isolate the predictive contribution beyond recipient-specific mode and mean using dose-certified target replacements.

Learning from disagreement. Human label variation has been argued to be signal rather than noise (Aroyo and Welty, 2015; Pavlick and Kwiatkowski, 2019; Uma et al., 2021; Plank, 2022; Basile et al., 2021), with ChaosNLI (Nie et al., 2020) and its distributional prediction follow-up (Zhou et al., 2022) as the reference nominal benchmark and LeWiDi (Leonardelli et al., 2023; Leonardelli et al., 2025) as the shared-task setting; our panels draw on POPQUORN (Pei and Jurgens, 2023), EmoBank (Buechel and Hahn, 2017), the Conversational Sarcasm Corpus distributed through LeWiDi-2025 (Jang & Frassinelli; Leonardelli et al., 2025), DICES (Aroyo et al., 2023), SummEval (Fabbri et al., 2021), HelpSteer2 (Wang et al., 2024) and the Buse–Weimer and Scalabrino readability studies. Recent work reports that soft labels beat majority labels on ChaosNLI, POPQUORN and CIFAR-10H (Singh et al., 2025) and that, on ChaosNLI, soft labels carry item-specific information that label smoothing cannot reproduce, with metric-dependent saturation in the number of annotations (Kohli, 2026). These results establish that the item-to-histogram association matters; they do not decompose which part of it does. Per-annotator and perspectivist models (Davani et al., 2022; Gordon et al., 2022; Fleisig et al., 2023) condition on rater identity; on DICES, our identity-conditioned heads changed predicted severity without improving held-out RPS (T6), a single-panel observation.

Constructed ordinal targets. SORD (Díaz and Marathe, 2019) and ordinal label-distribution learning (Wen et al., 2023) build soft targets from a hard label and label distances; label smoothing (Szegedy et al., 2016; Müller et al., 2019) spreads mass uniformly. We use such targets as reference arms, not as proposals: our question is whether the empirical residual allocation carries predictive value beyond what location-matched constructions provide.

Randomisation controls for supervision. The closest precedents intervene on the label side. Gheibi and Ghazizadeh (2026) use a shuffled stochastic-label-sampling control that breaks the example-to-distribution match while preserving dataset-level disagreement; Behrens and Zdeborová (2026) reassign teacher soft labels within classes in a distillation setting (appendix control). Label randomisation (Zhang et al., 2017) and permutation-based importance (Breiman, 2001; Fisher et al., 2019; Ojala and Garriga, 2010, who permute within classes) are evaluation-time or memorisation analogies. Our intervention differs in retaining recipient-specific statistics of each target — the modal class, or the exact mode and mean via a certified KL projection — and in reporting support and realised dose, which conditional permutation importance has long recognised as limiting when conditioning cells are small (Strobl et al., 2008; Debeer and Strobl, 2020).

Proper scoring and calibration. We score with the ranked probability score (Epstein, 1969; Gneiting and Raftery, 2007) and report calibrated sensitivities with a temperature-plus-climatology map that extends

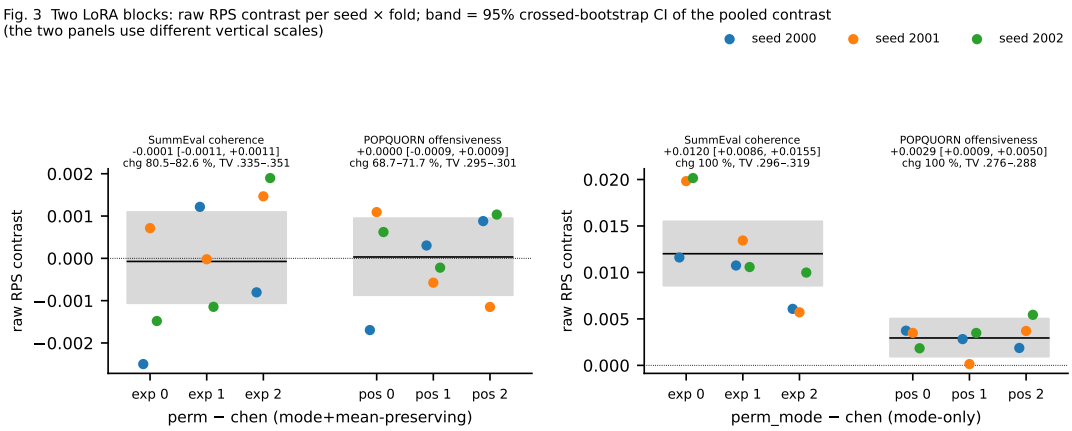


Figure 3: SummEval coherence LoRA replication block. Points are the nine seed-by-expert-fold raw RPS contrasts (three optimizer seeds, each paired with one item-assignment seed; seed-wise horizontal jitter). The band is a crossed-bootstrap 95% interval for their pooled contrast, resampling optimizer–assignment clusters and test documents; it is not a range of cell effects. Left: mode-and-mean-preserving replacement (perm − chen; 80.5–82.6% of targets changed, TV among changed 0.335–0.351). Right: mode-only replacement (perm_mode − chen; 100% changed, TV 0.296–0.319). Panels use different vertical scales.

temperature scaling (Guo et al., 2017); equivalence-band sensitivities follow the two-one-sided-tests convention (Schuirmann, 1987; Lakens, 2017) but are post hoc here.

3 Audit design

3.1 Interventions

For training item i with histogram h_i (K ordinal levels), mode z_i, mean μ_i:

- **soft** — h_i (Chen-style distributional target).
- **hard** — one-hot at z_i.
- **pmode** (mode-only donor replacement) — a maximum-cardinality assignment within mode cells such that every changed item receives a different histogram class with the same mode. It disrupts item-specific location and residual allocation beyond the mode, but the constrained derangement does not make donor histograms statistically independent of recipient histograms.
- **pmm** matches donors within mode and mean-bin cells and can retain small recipient–donor mean differences; **pmmx** additionally projects the donor to the recipient's exact mode and mean (deterministic KKT active-set KL projection, constraint- and optimality-certified to 1e-6, relative to the 1e-9-floored base). These interventions disrupt the item-specific association with the remaining allocation; only pmmx is generically exact mode-and-mean preserving.
- References: proto (mode-conditional prototype), mean2 (prototype KL-projected to the item mean under the mode constraint), cond (cross-fitted $\mathbb{E}[h \mid z, \text{mean-bin}]$), sordm/sordz (SORD Gaussians), shrinkage controls. Nominal (ChaosNLI): zc (modal class mass c, residual uniform-capped), pzc (donor residual allocation rescaled to c; certified $q_z = c$, $q_k \leq c$), pmode.

3.2 Dose and support

Support is the fraction of items in cells containing at least two histogram classes and is an upper bound on the changed fraction; the maximum-cardinality partial bijection may attain or fall below it. We report support, realised changed fraction, TV among changed items and TV over all items for every permutation arm (Table 1;

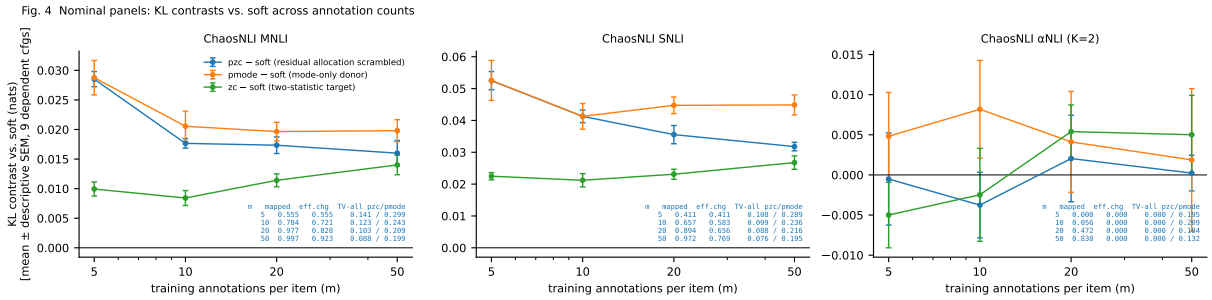


Figure 4: ChaosNLI nominal panels (MNLI, SNLI, αNLI; on αNLI the pzc intervention has zero final dose by construction): mean test KL to a fixed 50-annotation reference panel, contrasts vs. training on the m-annotation histogram, for residual-allocation scrambling with modal class and mass preserved (pzc), mode-only donor replacement (pmode) and the two-statistic target (zc); means and descriptive SEMs over nine dependent configurations. Blue text: mapped fraction, an effective-change sensitivity (TV $> 10^{-6}$) and the final TV over all items. A descriptive sensitivity to nested annotation-subsample size, not a causal annotation-count curve.

full certificates in Appendix B). In the three-to-five-rating panels many mode-and-mean cells contain a single histogram class — a dataset-specific support result, not a general combinatorial claim — so HelpSteer2 and EmoBank have low realised coverage (pmmx changes 0.280 and 0.135 of training items; EmoBank's cell support is 0.846 but within-cell class imbalance leaves few feasible replacements), SummEval, POPQUORN offensiveness and CSC sarcasm combine broad coverage with moderate TV, and Buse–Weimer and the two DICES panels change most items but by small TV, so their near-zero contrasts do not test large residual-shape changes.

3.3 Estimands

Raw RPS against held-out annotation positions (T = 1, $\lambda = 0$) is primary. Two calibrated sensitivities: a common pooled map (fixed roster) and arm-specific maps; the climatology weight hit the $\lambda \leq .5$ grid maximum in about half of the SummEval refits, so calibrated contrasts are reported with an extended-grid ($\lambda \leq .9$) sensitivity; the common-map results are grid-robust, the arm-specific ones are not. For the SummEval replication block, the raw-score interval is a crossed bootstrap over optimizer–assignment clusters and test documents; calibration-document resampling and map refitting apply only to the calibrated sensitivities, which are reported separately. Folds, split, evaluator set and base family remain fixed. Seed- and fold-specific estimates accompany every pooled interval. Heads ± values are descriptive SEMs over dependent configurations.

3.4 Pipelines

- Frozen-feature heads (the multi-panel instrument; claims scoped to frozen-feature learners): Qwen2.5-7B-Instruct last-token features (Qwen2.5-Coder-7B-Instruct for the code-readability panels), PCA-64 (16 for small panels), MLP, KL to target, early stopping on validation items vs training annotation positions; 5 master RNG streams × 3 group-disjoint item splits × 2 annotation-position splits (HelpSteer2 is grouped by prompt, SummEval by source document, CSC sarcasm by context, and the remaining panels by item); held-out annotation positions, which correspond to held-out raters only where persistent annotator identities are available (Buse–Weimer, DICES, Scalabrino, SummEval experts).
- LoRA judges form a separate end-to-end study on SummEval coherence: Qwen2.5-7B-Instruct LoRA, the Chen et al. hybrid objective 0.8·KL(p‖q) + 0.2·CE(mode(p), q) with target-distribution adversarial perturbation (chen), matched no-adversary control (chen0), arms as above; expert folds; split 0.

4 Results: frozen-feature heads

Setup. Ten ordinal panels (Table 1), 30 configurations each (5 RNG streams × 3 group-disjoint item splits × 2 annotation-position splits), MLP heads on PCA-reduced frozen features, KL to the target, RPS against

Table 1: Frozen-feature heads on ten ordinal panels: observed RPS contrasts (arm − soft; mean ± descriptive SEM over 30 dependent configurations, not inferential intervals). Columns pmmx/pmode/hard/sordm/cond give arm − soft. Dose = realised post-projection changed fraction of training targets under exact mode-and-mean-preserving replacement (pmmx; changed = TV $> 10^{-9}$) and the resulting TV averaged over all training items. pmode = mode-only replacement; sordm = SORD Gaussian at the item mean; cond = cross-fitted conditional reference.

panel	n_r	changed	TV-all	pmmx	pmode	hard	sordm	cond
SummEval coherence	8	0.73	0.27	-0.0010 ± 0.0003	$+0.0050 \pm 0.0006$	$+0.0072 \pm 0.0006$	-0.0053 ± 0.0010	-0.0012 ± 0.0003
POPQUORN offensiveness	8	0.64	0.22	$+0.0002 \pm 0.0002$	$+0.0068 \pm 0.0003$	$+0.0042 \pm 0.0003$	$+0.0159 \pm 0.0005$	-0.0002 ± 0.0001
CSC sarcasm (K=6)	6	0.70	0.29	$+0.0002 \pm 0.0003$	$+0.0022 \pm 0.0005$	$+0.0046 \pm 0.0006$	$+0.0123 \pm 0.0007$	-0.0003 ± 0.0004
Scalabrino readability	9	0.51	0.17	-0.0003 ± 0.0007	$+0.0028 \pm 0.0010$	$+0.0044 \pm 0.0012$	-0.0033 ± 0.0008	$+0.0001 \pm 0.0006$
POPQUORN politeness	5	0.42	0.21	$+0.0002 \pm 0.0002$	$+0.0033 \pm 0.0002$	$+0.0049 \pm 0.0002$	$+0.0026 \pm 0.0002$	-0.0002 ± 0.0002
HelpSteer2 helpfulness	5	0.28	0.19	$+0.0005 \pm 0.0001$	$+0.0006 \pm 0.0002$	$+0.0049 \pm 0.0003$	$+0.0088 \pm 0.0003$	$+0.0000 \pm 0.0001$
EmoBank valence	5	0.14	0.07	$+0.0006 \pm 0.0001$	$+0.0048 \pm 0.0002$	$+0.0042 \pm 0.0002$	$+0.0037 \pm 0.0004$	$+0.0005 \pm 0.0001$
Buse–Weimer readability	121	0.76	0.07	-0.0024 ± 0.0007	-0.0017 ± 0.0010	-0.0022 ± 0.0012	-0.0043 ± 0.0010	-0.0026 ± 0.0008
DICES-350	104	0.99	0.04	-0.0018 ± 0.0007	$+0.0022 \pm 0.0011$	$+0.0045 \pm 0.0014$	$+0.0429 \pm 0.0011$	-0.0016 ± 0.0007
DICES-990	60	0.98	0.04	-0.0001 ± 0.0002	$+0.0043 \pm 0.0002$	$+0.0108 \pm 0.0004$	$+0.0552 \pm 0.0012$	-0.0002 ± 0.0002

held-out annotation positions. Contrasts are arm − soft on the same configuration; ± values are descriptive SEMs over the 30 dependent configurations, not inferential intervals.

Support and dose. The mode-and-mean-preserving replacement changed between 0.135 (EmoBank) and 0.761 / 0.990 / 0.985 (Buse–Weimer / DICES-350 / DICES-990; post-projection change defined as TV $> 10^{-9}$) of training items, with support 0.46–1.00 (Table 1 and Appendix B). TV among changed items was 0.67 on the 3-rating HelpSteer2 panel, 0.50 on EmoBank and POPQUORN politeness, 0.41 on CSC sarcasm and 0.34 on POPQUORN offensiveness but only 0.09 / 0.04 / 0.04 on the three dense panels, so TV over all items ranged from 0.04 (DICES) to 0.27–0.29 (SummEval, CSC). Mode-only replacement changed 0.991–1.000 of items, with TV among changed items of 0.135–0.390 and maximum recipient–donor mean shifts up to 2.40 levels. Histogram-class matching produced no zero-TV donor pairs before projection; post-projection pmmx can nevertheless coincide with the original target, and on DICES-990 the certificates record 102 such pairs across the 18 dose-audit configurations.

Mode-and-mean-preserving replacement. Across the ten panels the exact variant (pmmx) produced observed pmmx − soft RPS contrasts from −0.0024 to +0.0006: SummEval −0.0010±0.0003, EmoBank +0.0006±0.0001, POPQUORN offensiveness +0.0002±0.0002, POPQUORN politeness +0.0002±0.0002, CSC sarcasm +0.0002±0.0003, DICES-990 −0.0001±0.0002, Scalabrino −0.0003±0.0007, HelpSteer2 +0.0005±0.0001, Buse–Weimer −0.0024±0.0007, DICES-350 −0.0018±0.0007. KL contrasts were likewise small (−0.018 to +0.009 nats). These observed contrasts show no average degradation larger than +0.0006 in the evaluated configurations; the descriptive SEMs do not constitute equivalence tests; their interpretation is limited by the low changed fractions on HelpSteer2 and EmoBank and by small perturbations on Buse–Weimer and the DICES panels. SummEval, POPQUORN offensiveness and CSC sarcasm provide the strongest ordinal head interventions because they combine broad realised coverage with moderate TV; on SummEval, pmmx changed 0.728 of items at TV 0.378 among changed items and produced an RPS contrast of −0.0010 and +0.004 KL. The mean-bin variant (pmm) was within 0.0006 of pmmx on nine panels and 0.0013 on DICES-350. Among panels where D-partial is defined, pmmx − soft ranged from −0.050 on Scalabrino and −0.026 on Buse–Weimer to +0.004 on POPQUORN offensiveness.

Mode-only replacement. Disrupting item-specific location as well produced positive observed contrasts on POPQUORN offensiveness (+0.0068±0.0003), SummEval coherence (+0.0050±0.0006), EmoBank (+0.0048±0.0002), DICES-990 (+0.0043±0.0002), POPQUORN politeness (+0.0033±0.0002), Scalabrino (+0.0028±0.0010) and CSC sarcasm (+0.0022±0.0005); HelpSteer2 (+0.0006±0.0002) and Buse–Weimer (−0.0017±0.0010) were small, and DICES-350 changed sign across otherwise comparable RNG streams (−0.0015±0.0008, +0.0022±0.0011, +0.0039±0.0014), which we report together without inferring a direction. The configuration SEMs do not capture this between-stream variability. These results show panel heterogeneity and are consistent with, but do not identify, the predictive value of item-specific location; the prespecified G × R check (T7, Appendix C) did not confirm the explanation on the held-out panels.

Constructed targets. No constructed target had the lowest observed RPS on every ordinal panel. The one-hot target (hard) had higher observed mean RPS than soft on nine panels (+0.0042 to +0.0108) and lower on Buse–Weimer (−0.0022). The SORD Gaussian centred on the item mean (sordm) had lower observed mean RPS

Table 2: Nominal ChaosNLI panels (frozen-feature heads): mean test KL to a fixed 50-annotation reference panel, contrasts vs. training on the m-annotation histogram (mean ± descriptive SEM over nine dependent configurations). Columns pzc/pmode/zc/cond give arm − soft in KL. pzc scrambles residual class allocation while retaining the modal class and its mass; dose = donor-mapped fraction and final TV over all training items. Binary αNLI (zero-dose limit) is reported in the text.

panel	m	mapped	TV-all	pzc	pmode	zc	cond
MNLI	5	0.555	0.141	$+0.029 \pm 0.001$	$+0.029 \pm 0.003$	$+0.010 \pm 0.001$	$+0.007 \pm 0.001$
MNLI	10	0.784	0.123	$+0.018 \pm 0.001$	$+0.021 \pm 0.003$	$+0.008 \pm 0.001$	$+0.006 \pm 0.001$
MNLI	20	0.977	0.103	$+0.017 \pm 0.001$	$+0.020 \pm 0.002$	$+0.011 \pm 0.001$	$+0.012 \pm 0.001$
MNLI	50	0.997	0.088	$+0.016 \pm 0.002$	$+0.020 \pm 0.002$	$+0.014 \pm 0.002$	$+0.013 \pm 0.001$
SNLI	5	0.411	0.108	$+0.052 \pm 0.003$	$+0.053 \pm 0.006$	$+0.022 \pm 0.001$	$+0.021 \pm 0.002$
SNLI	10	0.657	0.099	$+0.041 \pm 0.002$	$+0.041 \pm 0.004$	$+0.021 \pm 0.002$	$+0.017 \pm 0.003$
SNLI	20	0.894	0.088	$+0.036 \pm 0.003$	$+0.045 \pm 0.003$	$+0.023 \pm 0.002$	$+0.021 \pm 0.002$
SNLI	50	0.972	0.076	$+0.032 \pm 0.001$	$+0.045 \pm 0.003$	$+0.027 \pm 0.002$	$+0.024 \pm 0.001$

than soft on Buse–Weimer (−0.0043), Scalabrino (−0.0033) and SummEval (−0.0053) and higher on the other seven panels, severely so on DICES-350 (+0.0429), DICES-990 (+0.0552), POPQUORN offensiveness (+0.0159) and CSC sarcasm (+0.0123). The cross-fitted conditional reference (cond) was within ±0.0026 of soft on every panel, with lower observed mean RPS on seven of ten; the constrained projection of the prototype (mean2) was worse than cond on eight panels. Shrinkage controls were within ±0.0024 of soft everywhere. On Buse–Weimer, soft had higher observed RPS than several compressed targets but higher conditional disagreement ranking (D-partial 0.150 versus 0.101, 0.029 and 0.088 for hard, proto and sordm), an observed metric trade-off rather than an identified mechanism.

Semi-synthetic sanity check (T4). In a semi-synthetic world where item-linked shape was injected consistently for training and reference annotation positions — applied separately within each realised panel so that every item's mode and mean are preserved within the training panel and within the evaluation panel — the chosen feature-aligned injection was learnable by the heads in the observed configurations on Buse–Weimer: the soft − mean2 disagreement gap was +0.27±0.04 against an oracle gap of +0.57±0.02 (learnable fraction 0.47), and mean-bin donor replacement (pmm, mean-bin preserving) reduced disagreement recovery by −0.23±0.04. This demonstrates sensitivity to this injected signal, not to arbitrary residual-allocation effects. Under the same subset-wise injection the Scalabrino oracle gap was +0.02±0.02 and the POPQUORN offensiveness oracle gap −0.04±0.00 (the chosen injection is not recoverable from the reference positions there), and CSC sarcasm has a single held-out position so disagreement recovery is undefined; those panels supply no positive-control evidence, and the sanity check therefore rests on Buse–Weimer alone. (Earlier versions of this check injected on the full annotation row before the position split and are superseded.)

Nominal panels (ChaosNLI MNLI, SNLI and αNLI; Table 2). With m ∈ {5, 10, 20, 50} training annotations per item and KL to a fixed 50-annotation reference panel, scrambling the residual class allocation while retaining the modal class and its mass (pzc; constraint-certified, 0 violations) increased mean test KL at every m: pzc − soft from +0.029±0.001 (m = 5; mapped 0.555, support 0.79, final TV-all 0.141) to +0.018 (m = 10; mapped 0.784), +0.017 (m = 20; 0.977) and +0.016±0.002 (m = 50; 0.997, final TV-all 0.088). Some mapped targets differ from the original histogram only at the numerical-floor scale after rescaling, so a 'final changed fraction' depends on the declared tolerance; we report the mapped fraction and the final TV over all items as the primary dose statistics (Table 3). The two-statistic target zc had 0.008–0.014 higher KL than soft, and the cross-fitted conditional reference 0.006–0.013 higher; residual-shrinkage was within ±0.002. Residual-rank recovery moved in the same direction (soft − zc +0.055 to +0.064; pzc − soft −0.052 to −0.134). These are descriptive results over nine dependent configurations. On SNLI (1,514 items) the same pattern was larger: pzc − soft +0.052±0.003 (m = 5; mapped 0.411, final TV-all 0.108) to +0.032±0.001 (m = 50; mapped 0.972, TV-all 0.076), pmode − soft +0.041 to +0.053, zc 0.021–0.027 and cond 0.017–0.024 worse than soft, residual-rank recovery pzc − soft −0.13 to −0.23 (descriptive over nine dependent configurations per panel; the two panels share the annotation protocol but not items). The binary αNLI panel (K = 2) is the degenerate limit of this construction: retaining the modal class and its mass determines the whole two-class distribution, so the pzc target coincides with the original histogram (final TV over all items 0.000 at every m, although the mapped fractions were 0.000 / 0.056 / 0.472 / 0.830 at m = 5 / 10 / 20 / 50); the observed pzc − soft contrasts were −0.001 / −0.004 / +0.002 / +0.000, small relative to their descriptive SEMs of 0.002–0.006. Because arms use separate optimisation randomness, a zero-dose target need not give an exactly zero fitted-model contrast;

Table 3: LoRA judges (Qwen2.5-7B-Instruct; three optimizer seeds, each paired with one item-assignment seed; three held-out folds; split 0). Raw RPS contrasts with 95% crossed-bootstrap intervals over the three joint seed clusters and test documents (SummEval) or test items (POPQUORN); calibrated sensitivities resample calibration units and refit maps. Seed-marginal raw means, perm − chen: SummEval -0.0007 / +0.0007 / -0.0002, POPQUORN -0.0002 / -0.0002 / +0.0005; perm_mode − chen: SummEval +0.0095 / +0.0130 / +0.0136, POPQUORN +0.0028 / +0.0024 / +0.0036. Dose: SummEval perm changed 80.5–82.6% of targets (TV among changed 0.335–0.351), POPQUORN 68.7–71.7% (0.295–0.301); perm_mode changed 100%. POPQUORN position folds hold out ratings, not persons; a rater-aware (item × user) bootstrap widens its intervals (Section 4).

contrast	raw	common map	arm-specific
SummEval coherence (3 persistent experts held out; document-level inference)			
perm − chen (mode+mean preserved)	−0.0001 [-0.0011, +0.0011]	+0.0000 [-0.0006, +0.0008]	+0.0003 [-0.0015, +0.0022]
perm_mode − chen (mode only)	+0.0120 [+0.0086, +0.0155]	+0.0077 [+0.0053, +0.0109]	+0.0084 [+0.0051, +0.0121]
perm_mode − perm	+0.0121 [+0.0091, +0.0150]	+0.0077 [+0.0053, +0.0107]	+0.0081 [+0.0050, +0.0115]
POPQUORN offensiveness (3 annotation positions held out; item-level inference conditional on the rater panel)			
perm − chen (mode+mean preserved)	+0.0000 [-0.0009, +0.0009]	+0.0000 [-0.0007, +0.0008]	−0.0000 [-0.0009, +0.0008]
perm_mode − chen (mode only)	+0.0029 [+0.0009, +0.0050]	+0.0039 [+0.0022, +0.0060]	+0.0034 [+0.0016, +0.0053]
perm_mode − perm	+0.0029 [+0.0009, +0.0050]	+0.0038 [+0.0021, +0.0059]	+0.0035 [+0.0017, +0.0053]

we report αNLI as an implementation and optimisation-noise check, not as a third nominal replication or as evidence of equivalence. Mode-only replacement on αNLI, which does change targets (TV over all items 0.13–0.21), gave +0.002 to +0.008. The contrast with the ordinal panels is descriptive: on these three-class nominal tasks (MNLI, SNLI), scrambling mass beyond the modal class increased observed mean test KL under this learner and intervention, whereas the ordinal mode-and-mean-preserving replacements produced small observed contrasts at the doses available.

5 Results: LoRA judges

Setup. Qwen2.5-7B-Instruct with LoRA, the Chen et al. hybrid objective with target-distribution adversarial perturbation (chen). The three-seed replication roster comprised chen, mode-and-mean-preserving perm and mode-only perm_mode; hard, mean2 and proto were auxiliary checks conditional on the seed-2000 optimizer realisation, and chen0 is not part of the reported block. Both blocks use split 0 (60/10/30), three joint optimizer–assignment realisations and the roster chen / perm / perm_mode; raw RPS is primary, with common-map and arm-specific calibration as sensitivities. SummEval uses document-grouped splits, three persistent-expert folds and document-level inference; POPQUORN uses three randomly assigned annotation-position folds and item-level inference conditional on the realised finite rater panel.

SummEval coherence (replication block, T3b). Three optimizer seeds each paired with one item-assignment seed, across three fixed expert folds; the raw-score interval is a crossed bootstrap over the three joint optimizer–assignment clusters and test documents; calibration-document resampling and map refitting apply only to the calibrated sensitivities reported separately. Mode-and-mean-preserving replacement changed 80.5–82.6 % of training targets (TV among changed targets 0.335–0.351) and produced a near-zero raw contrast, −0.0001 [−0.0011, +0.0011]; its 90 % interval [−0.0009, +0.0009] lay within the ±.001, ±.002 and ±.003 sensitivity bands, all chosen post hoc (±.002 after observing the seed-2000 folds), which therefore cannot be presented as confirmatory bounds — exploratory evidence of a small average effect, not a confirmatory equivalence result — and the nine seed × fold cell estimates ranged from −0.0025 to +0.0019. Mode-only replacement increased raw RPS by +0.0120 (95 % crossed-bootstrap interval [+0.0086, +0.0155]), positive in all nine observed seed-by-fold cells (seed-marginal means +0.0095 / +0.0130 / +0.0136; fold-marginal means +0.0172 / +0.0116 / +0.0073). On the primary $\lambda \leq .5$ calibration grid (B = 10,000), common-map calibration attenuated this penalty (common +0.0077 [+0.0053, +0.0109]; common − raw −0.0045 [−0.0074, −0.0015]) and arm-specific calibration similarly (+0.0084 [+0.0051, +0.0121]; arm − raw −0.0039 [−0.0077, −0.0002]); $\lambda = .5$ was selected in 52.8 % of fit events (pooled over non-independent common-map and arm-specific-map fits). In the extended-grid sensitivity ($\lambda \leq .9$, B = 4,000; $\lambda = .9$ selected in 3.0 % of fit events) the common-map point estimate was unchanged (+0.0077; its interval [+0.0041, +0.0109] and common − raw −0.0049 [−0.0088, −0.0015] come from the separate B = 4,000 run) while the arm-specific estimate moved to +0.0097 [+0.0058, +0.0128] (arm − raw −0.0028 [−0.0067, +0.0004]); the arm-specific attenuation is therefore grid-sensitive. In a test-document

bootstrap conditional on one optimizer realisation, one-hot training had higher raw RPS than chen by +0.0186 [+0.0141, +0.0229]. With only three seed clusters, one split and one base family, these intervals are to be read cautiously.

POPQUORN offensiveness (second block, T3c). On 1,366 comments rated by eight annotation positions (819 / 137 / 410 items), mode-and-mean-preserving replacement changed 68.7–71.7 % of training targets at TV 0.295–0.301 among changed targets and produced a raw contrast of +0.0000 [−0.0009, +0.0009] (nine cells −0.0017 to +0.0011); common-map and arm-specific estimates were likewise near zero (+0.0000 [−0.0007, +0.0008]; −0.0000 [−0.0009, +0.0008]). Mode-only replacement changed every target and increased raw RPS by +0.0029 [+0.0009, +0.0050], positive in all nine observed cells (+0.0001 to +0.0054), and calibration did not attenuate it (common +0.0039 [+0.0022, +0.0060]). Its approximately fourfold smaller magnitude than SummEval's +0.0120 is descriptive; POPQUORN offensiveness is floor-heavy (58 % of ratings at the lowest level), but that does not identify the reason for the difference. The analysis script was the frozen SummEval script with only the panel, output root and resampling unit changed, committed before the first POPQUORN output. POPQUORN's loader assigns each item's ratings to positions and drops user identifiers, so its folds hold out a rating rather than a person: every test-evaluator user also rates training targets. This is not input leakage — the judge never sees identity — but the intervals above condition on the finite rater panel. A rater-aware pigeonhole bootstrap over items × users × seed clusters (raw estimand; 255 evaluator users) widened them to [−0.0014, +0.0015] for perm − chen and [−0.0004, +0.0065] (90 % [+0.0001, +0.0059]) for perm_mode − chen, so under rater-level resampling the POPQUORN mode-only penalty is directionally consistent but not distinguishable from zero at the 95 % level.

6 Discussion

The audit licenses statements about the average predictive consequence of a specific replacement under a specific learner and dose; it does not license "the learner ignores X" or "X is noise". Across the ordinal head panels the observed exact mode-and-mean-preserving contrasts ranged from −0.0024 to +0.0006, with coverage or TV limitations on several panels; the three panels that combine broad coverage with moderate TV (SummEval, POPQUORN offensiveness, CSC sarcasm) gave −0.0010, +0.0002 and +0.0002. On SummEval, mode-only replacement produced positive observed contrasts in both pipelines; positive head contrasts also appeared on both POPQUORN panels, DICES-990, Scalabrino and CSC sarcasm, but not uniformly. On ChaosNLI MNLI and SNLI the residual-allocation scrambling contrast was positive at every annotation count. The contrast between the ordinal panels and ChaosNLI motivates, but does not establish, a label-geometry explanation. Practically, the low support of three-to-five-rating cells limited the reach of this intervention without establishing that mode and mean generally determine a histogram or bound the value of full annotations; the ChaosNLI result cautions against discarding residual allocation on nominal tasks without task-specific validation; and no constructed two-statistic target is recommended, since none had the lowest observed RPS everywhere. The heads and the LoRA judge are separate instruments whose concordance on SummEval — small mode-and-mean-preserving contrasts and positive mode-only contrasts in both — is descriptive; the second LoRA block on POPQUORN offensiveness reproduced the same pattern with a smaller mode-only penalty. Methodologically, we recommend reporting intervention support, changed fraction, TV dose and separated raw and calibrated estimands in supervision-ablation studies; the vector-versus-histogram matching bug we caught illustrates why.

7 Limitations and scope

Frozen-feature heads and LoRA judges are different learners: head claims are scoped to frozen-feature learners. The LoRA pattern is observed on two ordinal datasets, but both blocks use one item split, one base family and only three joint optimizer–assignment realisations. POPQUORN folds hold out annotation positions rather than people even though persistent user identifiers are available; its item-bootstrap intervals therefore condition on the realised finite rater panel, and a rater-aware bootstrap widens them so that its mode-only interval includes zero. We claim equivalence nowhere: ±.001 was chosen post hoc on SummEval and was merely carried forward before inspecting POPQUORN outputs. Support and dose bound the interpretation of small contrasts: the exact replacement changed 28.0 % of HelpSteer2 items and 13.5 % of EmoBank items, Buse–Weimer and the DICES panels change most items but by small TV, and we claim equivalence nowhere — the primary raw SummEval 90 % interval [−0.00092, +0.00088] lies inside post hoc ±0.001, ±0.002 and ±0.003 bands, an exploratory

sensitivity only. Heads ± values are descriptive SEMs over dependent configurations and understate between-stream variability (DICES-350 mode-only sign changes across RNG streams); with only three seed clusters the SummEval intervals' stability and generalisation remain limited. Donor replacement under a deterministic class-flow rule does not produce histograms independent of the recipient and does not test alternative pairing rules or dose patterns; the projection and rescaling variants create new probability vectors through deterministic transformations under the retained constraints. Calibration is partly grid-dependent: extending the climatology grid from $\lambda \leq .5$ to $\lambda \leq .9$ left the common-map attenuation unchanged but moved the arm-specific estimate. HelpSteer2's five-rating subset is disagreement-selected and does not represent the full response population. Panels are not independent (the two SummEval criteria share documents; DICES-350 and DICES-990 share a collection), and the G × R check (Appendix C) is an exploratory diagnostic whose prespecified held-out form was not confirmed.

A Implementation audit and certificates

An earlier version matched donors on rater-indexed vectors, which paired items with identical histograms (zero-TV pairs: 18–23 % of SummEval LoRA perm pairs; on sparse head panels fewer than a third of nominally mapped examples received a different histogram). All reported numbers come from histogram-class matching with the zero-TV donor-pair count asserted to be 0 before projection; the vector-matched runs appear only as an implementation-audit row. Within each matching cell the numbers of assignments between histogram classes — hence the changed count and aggregate TV — are deterministic functions of the histogram-class identities and sizes; the permutation seed shuffles item identities within these slots (on SummEval 29–31 % of training items receive a different target between seeds) but not the class flow or dose, so each LoRA replication couples a new optimizer realisation with a new item-level assignment under the same class-flow rule and the three realisations cannot separate the two. Per-panel certificates (support, changed fraction, TV among changed items, TV over all items, zero-TV pairs, mean-shift range, post-projection dose) and the LoRA cell certificates are listed in the evidence manifest (results/MANIFEST.json), which records available hashes and reports absent ones as unavailable; for the nominal pzc target, zero donor TV is distinguished from zero post-rescaling TV.

B Exploratory G × R diagnostic (T7)

Held-out prespecified test (BW/DICES/HelpSteer2/Scalabrino): predicted sign in 2/4, Spearman 0.20; DICES not RNG-robust; HelpSteer2 a both-≈-0 case without an indifference region. The proposed location account did not survive the prespecified robustness checks; it is retained only as an exploratory diagnostic.

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